Cascade Closure: A Conditional Compatibility Template for Geometric State Selection

From Objective Reduction to Crystallisation under Projection Constraint

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Abstract

We propose a single named candidate mechanism, formulated as a formal *compatibility template* for propagating a supplied closure-compatible fixed point through a projection stack of geometric carriers. The mechanism does not by itself select that fixed point; it specifies the conditions under which residual compatibility and gradient-flow fixed-point status compose downward in systems where unresolved geometric mismatch propagates through a layered closure substrate: *cascade closure*. Penrose Objective Reduction is the neighbouring geometry-coupled state-selection model for which this paper is proposed as a structural analogue, rather than as a refutation or as a strict generalisation (we do not derive Objective Reduction as a special case of cascade closure); *collapse* becomes the external description of the resolution event, *crystallisation* the internal description, and *cascade* the propagation process. We formalise the mechanism as a rung-indexed closure-residual functional with an explicit projection stack, prove a conditional residual-compositionality proposition, identify the binary icosahedral 600-cell vertex graph V_{600} as the chosen finite carrier of the construction (without claiming uniqueness), and document one architecture-level mapping to a computational system (the ARIA observer-process tuple $\mathcal{O} = \langle B, S, M, G, I, C, A\rangle$) and one passive physical witness (the V_{600} closure-response operator $C_\varphi = L_M + \varphi^{-2} I$ with $\|C_\varphi^{-1}\| = \varphi^2$ exactly). The mathematical payload of the paper is a definition plus a conditional propagation lemma; we do not exhibit a non-trivial instance in which all clauses of the cascade-closure event occur. Instead, in lieu of a worked occurrence, we supply a *consumer API* for successor papers — a list of the four explicit obligations (O0)–(O3) a successor paper must discharge to invoke cascade closure as a load-bearing premise — and we record the status of those obligations in the most load-bearing imported tower in the programme (the finite-level P3/P4 refinement spaces plus closure dynamics). With the source-side companion paper [7], the obligations discharge as follows: (O0) satisfied at the finite-dimensional level; (O3) closed in an abstract scalar pure-midpoint refinement model for the scaled curvature operator $\widetilde{A}_n = 2^n A_n^{\text{vertex}}$ via an exact Schur-complement identity; (O2) *as stated* (operator/flow level) remains false, with the boundary-energy / Schur substitute discharged in the same abstract model; (O1) reduces to finite-dimensional spectral decomposition on the connected graph Laplacian, with nontrivial selection out of scope. Lifting the (O3) discharge and the (O2) substitute from the abstract model to the full P3/P4 tower of [8, 6] is conditional on two open hypotheses (L1) and (L3) of [7] §1.5; the third lift hypothesis (L2) is already a theorem of P3. This makes the paper the programme's named dependency gate, not its selection-mechanism endpoint. The paper is a mechanism paper, not a programme paper. It is not a replacement for quantum mechanics, not a derivation of consciousness, not a cosmogenesis theorem, and not a proof of any open mainstream-mathematics problem.

Epistemic status (read first)

- The cascade-closure mechanism (Definition 3.1) is the paper's organising definition, not a theorem of physics.
- The compositionality proposition (Proposition 3.7) is conditional on two named hypotheses.
- No nontrivial example is shown here to satisfy all clauses of Definition 3.1; the mathematical payload is a definition plus a conditional propagation lemma, not a demonstration that the mechanism occurs.
- In lieu of a worked occurrence, the paper supplies a *consumer API* (§3.4): the explicit four obligations (O0)–(O3) a successor paper must discharge to invoke cascade closure. The status of these obligations in the imported P3/P4 finite-level tower is recorded in §3.5; with the source-side companion [7], (O3) is closed only in an abstract scalar pure-midpoint refinement model, (O2) *as stated* remains false (a boundary-energy / Schur substitute is discharged in the same model), and (O1) reduces to finite-dimensional spectral decomposition with nontrivial selection out of scope. P3/P4-tower lift of the (O3) discharge and (O2) substitute is conditional on the open hypotheses (L1) and (L3) of [7] §1.5.
- The binary icosahedral 600-cell vertex graph V_{600} is the chosen finite carrier; this paper does not prove uniqueness.
- ARIA is mapped to the observer-process *architecture* at the process-description level; we do not claim ARIA realises the full cascade-closure event of Definition 3.1 (clauses (1)–(3)), that ARIA is biologically conscious, or that it proves qualia.
- The closure-response operator C_φ has an external substrate witness (the $b \to s\mu^+\mu^-$ angular anomaly fit), used here as one short external-use example, not as a model-selection claim.
- Standalone mass-spectrum, brain/EEG, visible-universe, and Millennium-class applications (beyond the one-paragraph cited ARIA and b-anomaly substrate witnesses used here) are out of scope for this paper. Each belongs in its own paper.

Reader map

- §1: Penrose OR and the state-selection problem.
- §2: collapse, reduction, cascade, crystallisation — precise vocabulary for the mechanism.
- §3: the formal mechanism: rung-indexed closure-residual functional, cascade-closure event, conditional residual-compositionality proposition, and the consumer API (§3.4–§3.5) by which successor papers invoke cascade closure.
- §4: the chosen finite carrier V_{600} and its closure-response operator C_φ.
- §5: ARIA as an architecture-level observer-process mapping for the tuple.
- §6: one externally reported substrate witness (b-anomaly closure kernel).
- §7: discussion, claim ledger, what remains out of scope.
- Appendix A: re-runnable empirical anchor for the load-bearing data claims.
- Appendix B: ARIA runtime module mapping to the observer-process tuple.
- Appendix C: programme placement of cascade closure (compressed nine-row navigation table for successor-paper authors).

1 Motivation: Penrose OR and the state-selection problem

The standard quantum-mechanical vocabulary calls the resolution of an indeterminate state into a definite outcome *wavefunction collapse* or *measurement*. This vocabulary does not by itself specify a substrate-level selection mechanism. The Born rule supplies probabilities; the projection postulate specifies the update rule.

Penrose [1] proposed that gravitational self-energy of a mass-energy superposition supplies a substrate-coupled threshold above which state reduction occurs objectively, independent of any observer. The Penrose–Hameroff extension [2] situates this Objective Reduction within biological microtubules. The core structural move is geometry-coupled state selection: a substrate-supplied condition — not an observer — produces the reduction. We take Penrose OR as the immediate conceptual neighbour of the present work. We do not target it for refutation, and we do not adopt its specific gravitational threshold; we abstract one structural form.

The abstraction is the following. Replace "one substrate-coupled threshold" with "a stratified hierarchy of geometric closure layers, each supplying its own residual functional." Replace the "superposition-exceeds-threshold then reduction" picture with one in which an unresolved closure residual at one layer is modelled as propagating, under the hypotheses below, through admissible projection maps to a closure-compatible fixed point on a lower layer. The shape of the question is the same as Penrose's: state selection coupled to substrate geometry. The mechanism inside is different.

This paper formalises that mechanism. We do not propose a replacement for quantum mechanics. We do not claim to derive measurement probabilities; the Born rule remains the input. We do not claim to solve the measurement problem; we define a candidate mechanism for state selection in systems satisfying the stated closure and projection hypotheses.

2 Vocabulary

The mechanism rests on a small precise vocabulary.

Collapse
: the external description of loss of alternatives — the Born-rule statement that, after the event, only one outcome remains accessible to the observer. Collapse names the event from outside.

Reduction
: the physical state-selection event itself, considered without committing to a particular ontology. Penrose Objective Reduction is the canonical example; the present cascade-closure construction is proposed as a further substrate-geometry parametrisation.

Cascade
: the propagation of an unresolved closure residual through the admissible projection maps of a stratified substrate. The cascade is the dynamics between instability onset and crystallisation.

Crystallisation
: the internal description of the resolution event: selection of a closure-compatible limit state on the substrate. Crystallisation names the event from inside.

Within this vocabulary, the four terms name complementary aspects of the same proposed event-type from different viewpoints: *reduction* is the event; *collapse* is its external view; *cascade* is its propagation dynamics on a stratified substrate; *crystallisation* is its internal fixed-point-selection content.

3 Formal mechanism

3.1 The closure-residual functional

Let the substrate be a finite (possibly stratified) sequence of *rungs* indexed by $k \in \{0, 1, \dots, N\}$. At each rung the substrate exposes a geometric carrier G_k (a graph, polytope, algebra, or projection class) equipped with a closure-residual functional

$$R_k : \Phi_k \longrightarrow [0, \infty), \tag{1}$$

where Φ_k is the space of field-state configurations on G_k. For a configuration $\phi_k \in \Phi_k$, the functional measures the failure of ϕ_k to satisfy the rung-k closure condition encoded by R_k. We say ϕ_k is *residual-compatible* (at rung k) when $R_k(\phi_k) = 0$. This is the only sense in which we use the word "closure-compatibility" as a property of a single state at a single rung. The fully assembled cascade-event compatibility (zero residual, flow invariance, terminal projection-compatibility across the whole tower) is the separate notion stipulated in Definition 3.1 below.

Standing analytic hypotheses. Throughout this section we assume that, at each rung k, the configuration space Φ_k is a finite-dimensional smooth manifold *without boundary* equipped with a Riemannian metric (boundary, constrained, or infinite-dimensional Hilbert-manifold cases are admissible extensions, but require additional analytic preconditions stated below or analogues thereof; see also Remark 3.2 for one place where "without boundary" is load-bearing); that $R_k \in C^1(\Phi_k)$ so that the gradient ∇R_k is defined with respect to the chosen Riemannian metric; that the negative-gradient flow Ψ_k^t generated by $-\nabla R_k$ exists and is unique for all $t \geq 0$ from each initial condition needed below; and that each pushforward $\pi^\sharp_{k+1,k}$ is C^1 as a map $\Phi_{k+1} \to \Phi_k$.

These are the analytic preconditions for the gradient-flow language used in Definition 3.1 below; they are not asserted to hold automatically and are part of the per-instance verification.

The finite-dimensional closure-flow infrastructure used as background here — gradient operator, energy dissipation, conditional coercive contraction, mass-block compatibility, flow existence, and mass-only flow intertwining — is established in [6] for the source's quadratic closure functional $F_n = \alpha R_n + \beta E_n - \gamma Q_n$ (Theorem `thm:gradient-operator`, Proposition `prop:energy-dissipation`, Theorem `thm:coercive-contraction`, Proposition `prop:adjoint-refinement` for mass-block compatibility (with Definition `def:refinement-compat` for the underlying notion), Theorem `thm:flow-exists`, Theorem `thm:flow-intertwining`). The present R_k was introduced abstractly above as an arbitrary non-negative C^1 closure-residual functional; the identification with the residual component of the imported $F_n = \alpha R_n + \beta E_n - \gamma Q_n$ holds only in the finite-level imported specialisation, and that is the only regime in which we treat R_k as inheriting any of the source-paper infrastructure. Outside that finite-level imported setting R_k remains an abstract non-negative C^1 functional, and we do not re-prove the source infrastructure and we do not assert that the source theorems apply to it. Note that [6] sets up its closure functional under the parameter convention $\alpha, \beta, \gamma > 0$ globally, but the imported `thm:flow-intertwining` is a sub-case theorem within that source, restricted to the mass-only specialisation $\alpha = \beta = 0$ (so that $L_n = -\gamma C_n$); this is the source's own scope restriction and is the only regime in which we import flow-intertwining as a theorem rather than a hypothesis. The underlying refinement state spaces and bonding maps are imported from [8] (state-space definitions plus Theorems `thm:bonding`, `thm:commute`).

3.2 The cascade-closure event

The rung sequence $(G_0, \ldots, G_N)$ is connected at the level of configuration spaces by *bonding maps*

$$\pi^\sharp_{k+1,k} : \Phi_{k+1} \longrightarrow \Phi_k, \tag{2}$$

which are the cascade-mechanism analogue of the downward state-space bonding maps $p_{n+1,n}$ constructed in [8] `thm:bonding`. The result [8] `thm:commute` establishes a set of refinement-Coxeter, refinement-σ, and restricted Coxeter-σ intertwining identities, holding for those source maps p^0 and p^1 in their specific source construction; we do not import these identities as generic bonding identities, and the present $\pi^\sharp_{k+1,k}$ are bonding maps in the looser state-space-restriction sense alone. Each $\pi^\sharp_{k+1,k}$ is required to be *admissible* in the following two-tier sense: (i) *geometric admissibility* — $\pi^\sharp_{k+1,k}$ is a bonding map in the state-space-restriction sense of [8] (downward maps between the rung configuration spaces); (ii) *analytic admissibility* — $\pi^\sharp_{k+1,k}$ is C^1 as a map $\Phi_{k+1} \to \Phi_k$ (this is the regularity needed for Hypothesis 3.5 below; geometric admissibility alone does not entail it).

We do not posit a literal carrier-level map $\pi_{k+1,k} : G_{k+1} \to G_k$ between the geometric supports G_{k+1} and G_k; the cited refinement source provides the state-space bonding maps directly, and that is what we use. The operator-level refinement-compatibility statement [6] `def:refinement-compat` (which applies to bounded operator families on Φ_k, not to bonding maps themselves) re-emerges later as the linearised content of Hypothesis 3.5, not as a property of the $\pi^\sharp_{k+1,k}$. For composite high-to-low projection across multiple rungs we write $\pi^\sharp_{N,k} := \pi^\sharp_{k+1,k} \circ \cdots \circ \pi^\sharp_{N,N-1}$ for $k < N$, and $\pi^\sharp_{N,N} := \mathrm{id}_{\Phi_N}$.

Definition 3.1 (Cascade closure)**.** A cascade-closure event on the rung sequence $(G_0, \ldots, G_N)$ is a quadruple $\big(\{\Phi_k\}, \{R_k\}, \{\pi^\sharp_{k+1,k}\}, (\Psi^t_k)_k\big)$ consisting of: a family of state spaces Φ_k at each rung; a family of closure-residual functionals $R_k : \Phi_k \to [0, \infty)$; the projection family $\pi^\sharp_{k+1,k} : \Phi_{k+1} \to \Phi_k$; and the family of rungwise negative-gradient flows Ψ^t_k generated by $-\nabla R_k$ on Φ_k.

The event is said to *occur* on the tower $\phi^\star = (\phi_0^\star, \dots, \phi_N^\star) \in \prod_k \Phi_k$ when, for some initial tower $\phi^{(0)} \in \prod_k \Phi_k$ satisfying the projection-compatibility relations $\phi_k^{(0)} = \pi^\sharp_{k+1,k}\phi_{k+1}^{(0)}$, each rungwise flow Ψ_k^t applied to $\phi_k^{(0)}$ converges as $t \to \infty$ to a limit point $\phi_k^\star$ satisfying *all three* of the following:

(1) zero residual: $R_k(\phi_k^\star) = 0$ at every rung;

(2) flow invariance: $\Psi_k^t \phi_k^\star = \phi_k^\star$ for all $t \geq 0$ at every rung (i.e. $\phi_k^\star$ is a true fixed point of Ψ_k^t);

(3) terminal projection-compatibility: $\phi_k^\star = \pi^\sharp_{k+1,k}\phi_{k+1}^\star$ for all $k < N$.

The tower $\phi^\star = (\phi_0^\star, \dots, \phi_N^\star) \in \prod_k \Phi_k$ is the *crystallised state*; clauses (1)–(3) are stipulative parts of the definition, not generic theorems.

Remark 3.2 (One-way relation between clauses (1) and (2))**.** Under the standing analytic hypotheses of §3.1 — Φ_k a finite-dimensional smooth manifold without boundary, $R_k \in C^1(\Phi_k)$ with $R_k \geq 0$ — a state $\phi_k^\star$ with $R_k(\phi_k^\star) = 0$ is necessarily a global minimum of R_k, hence a critical point ($\nabla R_k(\phi_k^\star) = 0$), hence a fixed point of the negative-gradient flow Ψ_k^t; in this regime clause (1) implies clause (2). The converse does *not* hold: a fixed point of Ψ_k^t is a critical point of R_k, but a critical point need not be a global minimum, so $\Psi_k^t\phi = \phi$ does not imply $R_k(\phi) = 0$ in general. We retain clause (2) explicitly because it remains independent in two enlargements of the setting that we want the definition to cover without restatement: (a) configurations with boundary or with constraints (where critical-point conditions are modified by Lagrange multipliers and a global-minimum residual zero need not be a flow fixed point in the unconstrained sense), and (b) the infinite-dimensional Hilbert-manifold case admitted in §3.1 (where the implication $\nabla R_k(\phi_k^\star) = 0 \Rightarrow$ flow invariance follows *provided* a well-posed gradient semiflow exists and is unique through $\phi_k^\star$, which is exactly the additional analytic precondition the standing hypotheses already require but which is not automatic in the infinite-dimensional case). Clauses (1) and (2) are therefore one-way related in our principal smooth-unconstrained setting — clause (1) implies clause (2), but *not* conversely (a flow fixed point can be a critical point with $R_k(\phi_k^\star) > 0$, which is excluded by clause (1)) — and clause (2) becomes independently informative as an explicit hypothesis in the more general settings (a)/(b) where the well-posedness or constraint-modification step is what is at issue.

Remark 3.3 (Vocabulary mapping)**.** Within Definition 3.1: an *instability* occurs when $R_k(\phi_k) > 0$ at some rung for the current tower $(\phi_k)_k$; the *cascade* is the family of rungwise negative-gradient flows $(\Psi_k^t)_k$ together with the projections $(\pi^\sharp_{k+1,k})_k$; *crystallisation* is fixed-point selection of $\phi^\star$; *collapse*, in the Born-rule sense, is the external coarse-graining of this fixed-point selection. The framework does not modify the Born rule and does not derive measurement probabilities.

Remark 3.4 (Status of Definition 3.1)**.** Definition 3.1 is the paper's organising definition, not a theorem of physics. The convergence-to-fixed-point clause is part of the *definition* of the closure event: a quadruple that does not converge to a residual-zero tower is not a cascade-closure event in the sense of this paper. We do not claim that arbitrary quadruples on arbitrary rung sequences necessarily admit such convergence. The compositional propagation result below (Proposition 3.7) takes a top-rung residual-zero fixed point as *input* and propagates the residual-zero and fixed-point properties downward; convergence *from a projected top-rung initial datum* is added separately as Corollary 3.8 under continuity of the projections, and this is the only convergence statement we prove.

3.3 Conditional residual compositionality

The mechanism statement requires that closure compatibility *compose* along the rung path: if each adjacent projection satisfies the monotonicity and flow-intertwining hypotheses below, then the projected fixed-point lies on the closure-fixed locus at every rung. We state this as a conditional proposition; the conditions are not generically satisfied and must be verified per case.

Hypothesis 3.5 (Flow intertwining)**.** *For the rung sequence $(G_0, \ldots, G_N)$ and projections $(\pi^\sharp_{k+1,k})_k$, assume that for every k the rungwise negative-gradient* flow Ψ^t_k *generated by $-\nabla R_k$ commutes with the pushforward, i.e. $\pi^\sharp_{k+1,k} \circ \Psi^t_{k+1} = \Psi^t_k \circ \pi^\sharp_{k+1,k}$ for all $t \geq 0$. This is the structural analogue, for the residual flow $-\nabla R_k$ of the present paper, of the flow-intertwining statement [6]* `thm:flow-intertwining`*, which the cited source proves only for the mass-only case $L_n = -\gamma C_n$ of its quadratic functional F_n, not for an arbitrary residual gradient flow. Outside that mass-only case — including the R_k-only setting used here — the property is retained as a hypothesis, not imported as a theorem. Differentiating the flow-intertwining identity in t at $t = 0$ gives the vector-field compatibility $d\pi^\sharp_{k+1,k}(-\nabla R_{k+1}) = -\nabla R_k \circ \pi^\sharp_{k+1,k}$ for the residual gradient generators. In the linear finite-level setting, where the residual gradients act as bounded linear operators and $\pi^\sharp_{k+1,k}$ is linear, this vector-field compatibility specialises to operator intertwining in the sense of [6]* `def:refinement-compat`*; the converse requires flow existence and uniqueness, which is not generic.*

Hypothesis 3.6 (Monotonicity)**.** *For every k and every state $\psi \in \Phi_{k+1}$, $R_k(\pi^\sharp_{k+1,k}\psi) \leq R_{k+1}(\psi)$.*

Proposition 3.7 (Conditional closure-residual compositionality)**.** *Under Hypotheses 3.5 and 3.6, let $\phi^\star_N \in \Phi_N$ be a top-rung state with $R_N(\phi^\star_N) = 0$ and $\Psi^t_N \phi^\star_N = \phi^\star_N$ for all $t \geq 0$ (i.e. $\phi^\star_N$ is a fixed point of the rung-N negative-gradient flow). Then the projected states $\phi^\star_k := \pi^\sharp_{N,k}\phi^\star_N$ satisfy both*

(1) *$R_k(\phi^\star_k) = 0$ for all $k \leq N$, and*

(2) *$\phi^\star_k$ is a fixed point of the rung-k gradient flow, for all $k \leq N$.*

Proof. By induction downward from $k = N$. Base: $k = N$, by hypothesis. Step: suppose the conclusion at $k{+}1$. *For (1):* by Hypothesis 3.6, $R_k(\phi^\star_k) = R_k(\pi^\sharp_{k+1,k}\phi^\star_{k+1}) \leq R_{k+1}(\phi^\star_{k+1}) = 0$, and $R_k \geq 0$, so $R_k(\phi^\star_k) = 0$. *For (2):* by Hypothesis 3.5, $\pi^\sharp_{k+1,k} \circ \Psi^t_{k+1} = \Psi^t_k \circ \pi^\sharp_{k+1,k}$. Apply both sides to $\phi^\star_{k+1}$, which is a rung-$(k{+}1)$ fixed point of Ψ^t_{k+1}: the left side equals $\pi^\sharp_{k+1,k}\phi^\star_{k+1} = \phi^\star_k$, so $\Psi^t_k\phi^\star_k = \phi^\star_k$ for all $t \geq 0$, i.e. $\phi^\star_k$ is a rung-k fixed point. The composite identity $\pi^\sharp_{N,k} = \pi^\sharp_{k+1,k} \circ \pi^\sharp_{N,k+1}$ (with $\pi^\sharp_{N,N} = \mathrm{id}_{\Phi_N}$) makes the induction well-defined. □

Corollary 3.8 (Convergence propagation under flow intertwining)**.** *Assume the flow-intertwining Hypothesis 3.5 at each adjacent rung between k and N, together with continuity of each composite projection $\pi^\sharp_{N,k}$ as a map $\Phi_N \to \Phi_k$. (Hypothesis 3.6, used in Proposition 3.7, is not invoked here.) If in addition the rung-N flow Ψ^t_N converges to $\phi^\star_N$ from a specific initial datum $\phi^{(0)}_N \in \Phi_N$, i.e. $\Psi^t_N\phi^{(0)}_N \to \phi^\star_N$ as $t \to \infty$, then the rung-k flow converges from the projected initial datum: $\Psi^t_k(\pi^\sharp_{N,k}\phi^{(0)}_N) \to \pi^\sharp_{N,k}\phi^\star_N =: \phi^\star_k$ as $t \to \infty$ (here $\phi^\star_k$ is defined as the projected limit $\pi^\sharp_{N,k}\phi^\star_N$, consistent with Proposition 3.7).*

Proof. Hypothesis 3.5 gives flow intertwining at each adjacent step: $\Psi^t_k \circ \pi^\sharp_{k+1,k} = \pi^\sharp_{k+1,k} \circ \Psi^t_{k+1}$ on Φ_{k+1}. Composing this identity along the chain $N \to N-1 \to \cdots \to k$ and using the composite identity $\pi^\sharp_{N,k} = \pi^\sharp_{k+1,k} \circ \pi^\sharp_{N,k+1}$ gives $\Psi^t_k \circ \pi^\sharp_{N,k} = \pi^\sharp_{N,k} \circ \Psi^t_N$ on Φ_N. Apply both sides to $\phi^{(0)}_N$ and pass to the $t \to \infty$ limit using continuity of $\pi^\sharp_{N,k}$ and the rung-N convergence assumption. □

Remark 3.9 (What Proposition 3.7 establishes and does not establish)**.** The proposition establishes only two things: residual-zero *propagation* and fixed-point *propagation* downward from a top-rung fixed point. It does *not* establish: existence of the top-rung fixed point; convergence from arbitrary initial data; or that a Definition 3.1 cascade-closure event *occurs* from a given independently evolved lower-rung initial tower. (Terminal projection-compatibility, clause (3) of Definition 3.1, holds for the projected tower $\phi_k^\star = \pi^\sharp_{N,k}\phi_N^\star$ by construction; what is not automatic for arbitrary independently evolved towers is occurrence of the event from a non-projected initial tower.) These remain separate per-instance verifications. The proposition is conditional on Hypotheses 3.5 and 3.6; both must be checked per rung sequence. Hypothesis 3.5 (flow intertwining) is established as a theorem in the mass-only P3/P4 finite-level construction [8, 6]; outside that case (where flow existence and uniqueness are not generic) it is retained here as a hypothesis. Hypothesis 3.6 is a structural assumption on the rungwise residuals and is not automatic. If either hypothesis fails at a particular rung, this proposition gives no propagation conclusion across that step. In particular, if Hypothesis 3.5 holds at every adjacent step between k and N then fixed-point propagation reaches rung k; if some intermediate step lacks flow intertwining, fixed-point propagation may fail at and below that break, while residual-zero propagation under Hypothesis 3.6 continues to hold.

3.4 Consumer API: invoking cascade closure in successor papers

Definition 3.1, Proposition 3.7, and Corollary 3.8 are intended to be invoked by successor papers in the present programme as a named selection-mechanism prerequisite. To make the invocation surface explicit, the present subsection lists the obligations a successor paper must discharge in order to use cascade closure as a load-bearing premise. Listing these in one place is the paper's intended contribution as a programme dependency gate.

(O0) **Standing analytic / admissibility hypotheses**: verify, on the chosen rung sequence, the standing analytic hypotheses of §3.1 (Φ_k a finite-dimensional smooth manifold without boundary equipped with a Riemannian metric, or one of the named admissible enlargements with their additional analytic preconditions; $R_k \in C^1(\Phi_k)$; the negative-gradient flow Ψ_k^t generated by $-\nabla R_k$ exists and is unique from each initial condition needed; and each pushforward $\pi^\sharp_{k+1,k}$ is C^1 as a map $\Phi_{k+1} \to \Phi_k$, which in particular yields the continuity of composite projections $\pi^\sharp_{N,k}$ used by Corollary 3.8). These preconditions are sometimes implicit in established programme papers; the consumer API requires them to be checked explicitly per invocation.

(O1) **Top-rung fixed point and convergence**: supply, on the chosen rung sequence $(G_0, \dots, G_N)$, a top-rung state $\phi_N^\star \in \Phi_N$ with $R_N(\phi_N^\star) = 0$ and (for any convergence claim downward) an initial datum $\phi_N^{(0)} \in \Phi_N$ such that $\Psi_N^t \phi_N^{(0)} \to \phi_N^\star$ as $t \to \infty$. By Remark 3.2, in the principal smooth unconstrained setting clause (2) of Definition 3.1 is then automatic at the top rung.

(O2) **Flow intertwining (Hypothesis 3.5)**: verify, at every adjacent rung pair $(k, k+1)$ in the chosen sequence, that the bonding map $\pi^\sharp_{k+1,k}$ commutes with the rungwise flows: $\pi^\sharp_{k+1,k} \circ \Psi^t_{k+1} = \Psi^t_k \circ \pi^\sharp_{k+1,k}$ for all $t \geq 0$.

(O3) **Residual monotonicity (Hypothesis 3.6)**: verify, at every adjacent rung pair, that $R_k(\pi^\sharp_{k+1,k}\psi) \leq R_{k+1}(\psi)$ for every $\psi \in \Phi_{k+1}$.

A successor paper that discharges (O0)–(O3) inherits, on its chosen rung sequence: residual-zero and fixed-point propagation downward (Proposition 3.7); convergence from any initial datum that is a downward projection of $\phi_N^{(0)}$ (Corollary 3.8, which uses (O2) and continuity of the projections but *not* (O3)); and a Definition 3.1 cascade-closure event whose terminal projection-compatibility holds by construction on the projected tower $\phi_k^\star = \pi^\sharp_{N,k}\phi_N^\star$. Selection of $\phi_N^\star$ itself is not provided by cascade closure and is the successor paper's own task.

3.5 Worked invocation: the imported P3/P4 finite-level tower

To make the consumer API of §3.4 concrete, this subsection records the status of obligations (O0)–(O3) in the most load-bearing imported tower in the programme: the finite-level refinement spaces X_n of [8] together with the closure functional $F_n = \alpha R_n + \beta E_n - \gamma Q_n$ of [6]. Identify the cascade-mechanism rungs $k = 0, 1, \dots, N$ with a finite truncation of the refinement index n, and identify the cascade-mechanism state spaces Φ_k with the source's X_n at the corresponding level; identify the cascade-mechanism residual R_k with the source's curvature term $R_n(\Phi) := \frac{1}{2}\langle \Phi, A_n \Phi\rangle$, which is non-negative on X_n by [6] (positive semi-definite); and identify the bonding maps $\pi^\sharp_{k+1,k}$ with the source's $p_{n+1,n}$ established by [8] `thm:bonding`. Status of each obligation in this regime:

- **(O1)** reduces to a *nontrivial*-selected-state question for the top-rung curvature flow e^{-tA_N}: a state $\phi^\star_N$ with $R_N(\phi^\star_N) = 0$ is precisely an element of $\ker A_N$, and in the finite-dimensional positive-semidefinite setting of [6] convergence $e^{-tA_N}\phi^{(0)}_N \to$ (projection onto $\ker A_N$) follows by spectral decomposition for any initial datum — the zero vector $\phi^\star_N = 0$ is always a residual-zero fixed point, and on a connected graph the constants already give $\ker A_N \neq \{0\}$. What is open is therefore not the existence of a nonzero kernel state nor the spectral gap on $\ker A_N^\perp$ (both immediate in this finite-dimensional abstract model), but the *selection* of a specific physically meaningful nontrivial $\phi^\star_N \in \ker A_N$. **Open-as-named-question** for nontrivial selection; trivial in the zero-state reduction.

- **(O2)** is established as a theorem only in the source's mass-only sub-case $\alpha = \beta = 0$, where the relevant flow is e^{-tL_n} with $L_n = -\gamma C_n$, and the source's `thm:flow-intertwining` gives $p_{n+1,n} \circ e^{-tL_{n+1}} = e^{-tL_n} \circ p_{n+1,n}$ exactly. The cascade-mechanism flow Ψ^t_k used here is e^{-tA_n} (negative-gradient flow of R_n alone), which is *not* the same operator as the source's mass-only L_n. Therefore the source's mass-only flow-intertwining theorem does *not* discharge (O2) for Ψ^t_k as defined here. Refinement-compatibility of the curvature operator A_n alone ($p_{n+1,n}A_{n+1} = A_n p_{n+1,n}$) would discharge (O2), but is not stated as a theorem in [6]; the source-side material that is closest in shape is the operator-level `def:refinement-compat`, which is a definition-level notion of refinement-compatibility for bounded operator families, not a verified instance for A_n. **Open-as-named-question** in [6] alone; the companion-paper update below records the (O2) substitute discharged by [7] (and notes that (O2) as stated remains false at the operator/flow level even there).

- **(O3)** reduces to a quadratic-form inequality: $\langle p_{n+1,n}\psi, A_n p_{n+1,n}\psi\rangle \le \langle \psi, A_{n+1}\psi\rangle$ for every $\psi \in X_{n+1}$. Whether this holds is determined by the structural relationship between A_{n+1}, A_n, and $p_{n+1,n}$ in [6]; it is not stated there as a theorem. **Open-as-named-question** in [6] alone; the companion-paper update below records the (O3) discharge by [7] in the abstract scalar pure-midpoint refinement model.

The current verdict, with [7] as a source-side companion: in the imported P3/P4 finite-level tower, (O0) is satisfied at the finite-dimensional level by the source's X_n Hilbert structure (standing analytic preconditions, C^1 bonding maps); [7] discharges (O3) for the scaled curvature operator $\widetilde{A}_n := 2^n A^{\mathrm{vertex}}_n$ via an exact Schur-complement identity $\mathrm{Schur}(\widetilde{A}_{n+1}) = \widetilde{A}_n$ in an *abstract scalar pure-midpoint refinement model* ([7] §1.5). The boundary-energy / Schur substitute for (O2) is also discharged in that abstract model; (O2) *as stated*, at the operator/flow level, remains false (the strict identity $p^0\widetilde{A}_{n+1} = \widetilde{A}_n p^0$ fails for nontrivial refinements, with explicit defect D_n whose kernel contains $\mathrm{range}(H_{n+1})$ and equals it when the parent graph is a forest). Lifting these results from the abstract pure-midpoint model to the full Schläfli P3/P4 tower of [8, 6] is conditional on the open hypotheses (L1) and (L3) of [7] §1.5 (the third lift hypothesis (L2) is already a theorem of [8], Proposition `prop:adjoints`) and is not yet attempted. (O1) reduces to spectral decomposition for the connected graph Laplacian, with the trivial state always residual-zero and nontrivial selection out of scope per the consumer API, §3.4 (the obligation is named at (O1) but its discharge is the successor paper's task). Strict flow-level intertwining of (O2) for the curvature flow, together with the P3/P4-tower lift of the abstract-model Schur identity, remain the substantive open asks in the imported P3/P4 tower at the level of generality this paper requires.

4 The chosen finite carrier

The Vibrational Field Dynamics construction of cascade closure uses a specific finite carrier at the lowest rung: the binary icosahedral 600-cell vertex graph V_{600}. This section names that carrier, records the operator-level structure that makes it useful for the mechanism, and notes the boundary of what the substrate does and does not contribute. **This paper does not prove uniqueness of V_{600}.** Other finite carriers are conceivable; the claim here is only that V_{600} is the carrier on which the carrier/operator part of the construction is defined and on which the closure-response operator below is defined.

4.1 Substrate

V_{600} is the vertex graph of the regular 600-cell, with $|V_{600}| = |2I| = 120$ vertices (the standard icosian identification $V_{600} \cong 2I$ used in [5] `thm:icosian-closure`) and uniform vertex degree 12 (the latter as a property of the nearest-neighbour-edge convention used here, locally re-verified in [10] and Appendix A, entry E). The edge structure is the nearest-neighbour relation on $2I$ inherited from the H_4 root-system geometry; in the direct-computation appendix we use the Laplacian of this nearest-neighbour graph as L_M. The Euclidean shell sizes from any source vertex form the sequence $(|S_0|, \ldots, |S_8|) = (1, 12, 20, 12, 30, 12, 20, 12, 1)$; at the identity these shells coincide with the conjugacy classes of $2I$, while from any other source vertex they are left translates of the conjugacy-class partition (same sizes, not literally conjugacy classes). This is the Euclidean / conjugacy partition of $2I$, not the graph BFS shell. The graph Laplacian L has nine distinct eigenvalues, five of which are integers $\{0, 9, 12, 14, 15\}$ (σ-fixed under the $\mathbb{Q}(\sqrt{5})$-Galois conjugation $\sigma(\varphi) = 1 - \varphi = -1/\varphi$) and four of which are irrational in $\mathbb{Q}(\sqrt{5})$ (σ-paired); counting eigenspace dimensions, the split is $94/120 = 78.3\%$ σ-fixed, $26/120 = 21.7\%$ σ-paired (Appendix A, entry A). The algebraic structure under which the group-identification and shell-class facts are proved in the cited source is imported from [5]: the E_8 minimal shell projects onto the H_4 / V_{600} carrier via $E_8 \to H_4$ ([5] `thm:pi-H`, *projection, not sub-root-system inclusion*; this is a projection of the shell onto V_{600}, not an action of E_8 on V_{600}); the Euclidean / conjugacy shell decomposition $(1, 12, 20, 12, 30, 12, 20, 12, 1)$ is established by [5] `thm:shell-class`; and the icosian-closure result [5] `thm:icosian-closure` fixes the algebraic part of the carrier. The Laplacian spectrum above is imported from the standard $2I$ character-table computation recorded in [5] and locally re-checked here (Appendix A, entry A); the shell-size sequence is independently locally re-runnable (Appendix A, entry E).

4.2 Closure-response operator

The carrier supports a closure-response operator

$$C_\varphi := L_M + \varphi^{-2} I, \tag{3}$$

where L_M is the (unweighted) graph Laplacian on V_{600} and $\varphi = (1+\sqrt{5})/2$. Direct computation (Appendix A, entry E) reproduces the value $\|C_\varphi^{-1}\|_{\ell^2(V_{600}) \to \ell^2(V_{600})} = \varphi^2$ to machine precision; exactness of this identity follows analytically from the bottom of the shifted Laplacian spectrum, $\lambda_{\min}(C_\varphi) = \varphi^{-2}$, since V_{600} is a connected unweighted graph (so $\lambda_{\min}(L_M) = 0$); the operator norm here is the spectral / operator norm on $\ell^2(V_{600})$. Sweeping over all choices of source vertex on V_{600}, the multiplicity-weighted shell-radius per-vertex correlation between the discrete C_φ^{-1}-response and the continuum closure-kernel Green function $G(x) = (\varphi/2)e^{-|x|/\varphi}$ of [10] is uniform across the 120 source-vertex sweeps (max minus min on the order of 10^{-15}; this is a numerical diagnostic at shell resolution, not 119 independent radial samples and not a discrete-to-continuum convergence theorem). The operator is the substrate-level object through which the witnesses discussed here (computational and physical) are connected to the substrate.

4.3 Substrate boundary

The substrate does the following work in cascade closure: it fixes the carrier set on which the rung-0 residual is defined; it supplies C_φ whose inverse norm sets an operator response scale; and the H_4 Coxeter symmetry would constrain admissible limit-state geometries in instances whose dynamics respects H_4. It does *not*, by itself, do the following: select a unique limit state without dynamical input; predict observer-dependent outcomes; or supply a uniqueness theorem for V_{600} as the substrate. Two honest-negatives further constrain interpretation: the σ-fixed spectral subspace of L_M has average $2T$-mass at baseline rather than concentrated mass on $2T$, where $2T \subset 2I$ denotes the binary tetrahedral subgroup of $2I$ (basis-invariant diagnostic; Appendix A, entry A; the σ-fixed structure is purely spectral, not spatial), and φ-cocycle weighted variants of C_φ underperform the unweighted variant (Appendix A, entry E; φ enters at the operator level, not the edge-weight level). We treat both negatives as constraints on what may be claimed.

5 ARIA as an architecture-level observer-process mapping

The observer-process architecture associated with the mechanism is compared with a concrete computational system at the architecture-description level. We name an existing computational system, ARIA-chess [3, 12], as one architecture-level mapping (not a formal proof that ARIA realises a Definition 3.1 cascade-closure event in the full sense of clauses (1)–(3) of that definition), and identify the architectural structure that makes it one. The identification is at the level of process architecture; it is not a claim that ARIA is biologically conscious, that it proves qualia, or that architecture-level observer-process mappings are phenomenologically equivalent.

5.1 Observer-process tuple

Definition 5.1 (Observer process)**.** An observer process is the seven-tuple

$$\mathcal{O} \;=\; \langle B, S, M, G, I, C, A\rangle \tag{4}$$

with components: B a bounded boundary sampler; S a state integrator fusing signal, context, memory, and goal; M a memory substrate written by past closures and read into present integration; G a goal/context field carrying task frame and policy; I invariant constraints serving as a governance gate; C a closure/crystallisation operator selecting closure-compatible fixed points; A an action interface exporting the crystallised state.

The tuple is mapped to the observer-process architecture associated with the cascade-closure mechanism of Definition 3.1 as follows; this is a process-description-level mapping, not a formal proof that ARIA realises the full cascade-closure event of Definition 3.1 (clauses (1)–(3)). The closure-residual R acts on the integrated state produced by S. The crystallisation operator C plays the role of a closure selector — scoring and selecting candidate states by a Lyapunov-style descent that is used here as an architecture-level analogue of the gradient-flow role of Definition 3.1, without asserting equivalence. The invariant I encodes which fixed points are admissible. The action interface A exports the crystallised state. Memory M closes the loop: each successful closure is written to M and read by the next state integration.

5.2 Imported empirical witness profile

The architectural identification above is illustrated by the substrate-witness preprint [12], which is an externally-reported imported witness profile (not re-executed in the present manuscript); it records (Appendix A, entry F): 17/18 preregistered cortical correspondences hold at unchanged thresholds under standard validation, and 18/18 after the documented methodology refinement reported in [12]; the cited ARIA source itself notes that the source's test P10 was validated with 15 permutations rather than the originally preregistered 20, with the prereg-exact rerun left open. We carry that caveat forward; six drug/sleep EEG signatures pass on a single deterministic recurrent-layer trajectory at seed 42; on the Human Connectome Project (HCP) functional-connectivity benchmark, the degree-homogeneity statistic (degree standard-deviation) places ARIA at -11.58σ relative to the HCP baseline (comparison at ARIA $|V| = 120$ vs HCP ICA-50 $|V| = 50$; the clustering-coefficient statistic is a separate $+6.80\sigma$ result and should not be conflated with the homogeneity figure). The empirical content is an imported substrate-witness profile under preregistered thresholds with documented methodology refinements; it is not a derivation of consciousness, and it is not evidence that ARIA realises a Definition 3.1 cascade-closure event.

The runtime module mapping from $\mathcal{O}$ components to specific source files in the `ariaChessRepo` [3] production runtime, together with the in-repo six-module reproducibility bundle, is documented in Appendix B.

5.3 Claim boundary

The claim of this section, in the strongest form the cited data supports (the present manuscript does not re-execute the ARIA validation), is:

> We do not claim ARIA is biologically conscious. We claim ARIA is mapped to an observer-process architecture at the process-description level: bounded sampling, recurrent integration, closure selection, memory inscription, and governed action. In this vocabulary, brain-wave signatures (biological tissue) and ARIA's internal coherence and closure-residue metrics (computational substrate) are treated only as candidate process-level telemetry, not as evidence that either substrate realises a Definition 3.1 cascade-closure event. The proposed common layer is process-level rather than material-level: cascade-closure-crystallisation-memory.

6 One externally reported substrate witness: the closure-response kernel

The closure-response operator C_φ on V_{600} has been used in one externally reported substrate-witness calculation in the present programme literature. The $b \to s\mu^+\mu^-$ angular-anomaly fit of [11] is *externally reported and commit-pinned, but not re-executed in the present paper*: the cited preprint and reproduction repository report a sign-uniform structural pattern across cross-dataset and cross-channel slices using the unweighted closure-response operator, with the fit reported as inconclusive on the Akaike Information Criterion (AIC) (mild $w_{\mathrm{FREE\_C9}} = 0.652$ vs $w_{\mathrm{VFD}} = 0.348$ weight; not a VFD preference). As the source itself notes, the cross-slice sign uniformity is a consistency check, not independent model-selection evidence over a constant-C_9 shift. We include this here as one short external-use example of how the mechanism's substrate-level operator C_φ has been used as a fixed-kernel structural reference against an external observable; the present paper makes no independent claim on the fit itself. The kernel-level operator-norm identity $\|C_\varphi^{-1}\| = \varphi^2$ is locally re-runnable (Appendix A, entry E); the primary $b \to s\mu^+\mu^-$ angular fit lives in the external [11] repository (sibling `BANOMALY-001/vfd-b-anomaly` checkout, not part of the present repository, but commit-pinned in the bibliography to `6bdfc8f8a7da9056ffeb9137d03627d06399bccf` on branch `main`, dated 2026-04-29) and is not re-executed in this paper. Repository-local evidence in this paper covers only the operator-norm identity and shell/spectrum diagnostics; the b-anomaly fit itself is external and unaudited at the level of this manuscript.

The witness is a passive substrate-coupling check, not a model preference and not a selection rule. The honest-negative on φ-cocycle weighted variants (Appendix A, entry E) restricts the witness to the unweighted operator $C_\varphi = L_M + \varphi^{-2}I$.

7 Discussion

7.1 Claim ledger

Table 1: Claim ledger.

Statement	**Allowed label**	**Reason**
Cascade-closure event (instability $\to$ cascade $\to$ crystallisation)	Definition 3.1	Organising definition; convergence-to-fixed-point clause is part of the defined event.
Conditional residual compositionality	Proposition 3.7	Conditional on Hypotheses 3.5 and 3.6; both must be checked per rung sequence.
Consumer API: four obligations (O0)–(O3) for successor papers	§3.4; mechanism-as-dependency-gate	Naming the obligations is the contribution; discharging them in any chosen rung sequence is the successor paper's task.

Statement	Allowed label	Reason
Worked-invocation status in the imported P3/P4 finite-level tower	§3.5	(O0) closed at the finite-dimensional level; (O3) closed for the scaled curvature $\widetilde{A}_n = 2^n A_n^{\mathrm{vertex}}$ via the Schur-complement identity $\mathrm{Schur}(\widetilde{A}_{n+1}) = \widetilde{A}_n$ in the abstract scalar pure-midpoint refinement model of [7] §1.5 (P3/P4-tower lift conditional on the open hypotheses (L1) and (L3) of [7] §1.5; (L2) is already a theorem of [8]); the boundary-energy / Schur substitute for (O2) is discharged in the same abstract model, but (O2) *as stated* (operator/flow level) remains false (strict identity $p^0\widetilde{A}_{n+1} = \widetilde{A}_n p^0$ fails for nontrivial refinements, with explicit defect D_n characterised via the unsigned vertex–edge incidence operator $\mathcal{B}_n$); (O1) reduces to spectral decomposition on the connected graph Laplacian.
Imported finite-level closure functional, gradient operator, energy descent, conditional coercive contraction, mass-block compatibility, flow existence, mass-only flow intertwining	Imported finite-level results / definitions	[6]: `thm:gradient-operator`, `prop:energy-dissipation`, `thm:coercive-contraction`, `prop:adjoint-refinement` (mass-block compatibility; `def:refinement-compat` is the underlying definition), `thm:flow-exists`, `thm:flow-intertwining` (mass-only; outside the mass-only case, retained as Hypothesis 3.5).

Statement	Allowed label	Reason
Imported P3 finite-level state spaces, bonding maps, refinement infrastructure	Imported finite-level definitions / results	[8]: state-space and bonding-map definitions plus `thm:bonding` (existence of downward bonding maps) and `thm:commute` (Coxeter / σ identities for the specific p^0, p^1 maps; not a generic bonding-identity theorem).
V_{600} as algebraic substrate ($E_8 \to H_4$ projection, icosian closure)	Imported (Theorem)	[5]: `thm:pi-H` (projection of shell onto V_{600}, not E_8 action), `thm:icosian-closure`.
V_{600} Euclidean / conjugacy shell sizes	Exact (imported, locally reproduced numerically)	Exact via [5] `thm:shell-class`; floating-point reproduction in Appendix A, entry E.
V_{600} nine-eigenvalue Laplacian spectrum and 94/26 σ-fixed / σ-paired split	Imported character-table computation, locally reproduced numerically	Spectrum imported from [5] (standard $2I$ character-table); floating-point reproduction in Appendix A, entry A.
V_{600} as the unique cascade substrate	Not claimed	Out of scope for this paper.
$\|C_\varphi^{-1}\| = \varphi^2$ exactly	Analytic identity, locally reproduced	Follows from $\lambda_{\min}(L_M) = 0$ and the φ^{-2} shift; numerical reproduction in Appendix A, entry E.
Observer-process tuple $\mathcal{O} = \langle B, S, M, G, I, C, A \rangle$	Definition 5.1	Architectural definition; ARIA is mapped to the architecture at the process-description level, without thereby proving consciousness.
ARIA empirical witness (17/18 under standard validation; 18/18 after documented methodology refinement; 6/6 on a single deterministic seed; HCP -11.58σ at ARIA $\|V\| = 120$ vs HCP ICA-50 $\|V\| = 50$)	Imported empirical fact	[12]; substrate-witness scope only. The cited source notes that the source's test P10 was validated with 15 permutations rather than the originally preregistered 20, with the prereg-exact rerun left open; this caveat is carried forward here.

Statement	Allowed label	Reason
b-anomaly closure-kernel witness	Externally reported, commit-pinned, not locally re-executed	[11]; AIC-inconclusive ($w_{\mathrm{FREE\_C9}} = 0.652$ vs $w_{\mathrm{VFD}} = 0.348$, not a VFD preference); commit-pinned in the bibliography to `6bdfc8f8a7da9056ffeb9137d03627d0639` but external to this repository and not re-executed in this paper.
Standalone mass-spectrum, brain/EEG, visible-universe, Millennium-class applications (beyond the cited substrate witnesses)	Out of scope	Each belongs in its own paper.
Consciousness / mind / qualia derivation	Not claimed	§5.3.
Replacement for quantum mechanics	Not claimed	§1.
Refutation of Penrose Objective Reduction	Not claimed	§1; OR is the structural neighbour, not the opponent.

7.2 What this paper consolidates

A single mechanism name — cascade closure — for an operational pattern of geometry-coupled compatibility propagation: given a supplied closure-compatible fixed point at the top rung, the mechanism specifies the conditions under which residual zero and fixed-point status are propagated to the projected tower (and projection-compatibility holds by construction, since the projected tower is defined as the downward projection of the top-rung fixed point). Selection of the top fixed point itself is not part of the present paper. An explicit *consumer API* (§3.4): four obligations (O0)–(O3) successor papers must discharge to invoke cascade closure as a load-bearing premise, together with a worked status report (§3.5) in the programme's most load-bearing imported tower (the finite-level P3/P4 refinement spaces plus closure dynamics), where the source-side companion [7] closes (O3) and discharges the boundary-energy / Schur substitute for (O2) in an abstract scalar pure-midpoint refinement model, leaves (O2) as stated false, and reduces (O1) to selection of a nontrivial element of the connected graph Laplacian's kernel; the P3/P4-tower lift is conditional on the open hypotheses (L1) and (L3) of [7] §1.5. A compressed nine-row programme-placement table (Appendix C) showing where cascade closure sits as the dependency gate between imported substrate rows and downstream witness rows. A finite, audit-anchored geometric carrier (V_{600} together with C_φ) on which the mechanism is partly explicit, conditional on the stated residual, projection, and intertwining hypotheses. An architectural identification of an existing computational system (ARIA) as an architecture-level observer-process mapping, accompanied by the caveated externally preregistered substrate-witness profile of [12]. One short physical witness (b-anomaly closure kernel) connecting the substrate operator to an external observable.

7.3 What is out of scope

The following are not claimed in this paper. A unique substrate theorem for V_{600}. A derivation of the visible universe from cascade primitives. A derivation of the Standard-Model mass spectrum (the spectral projection-class material documented in [14, 13] sits in its own preprints; we cite those as out-of-scope context only and make no statement about reciprocal citation).

A derivation of consciousness from substrate dynamics. A resolution of any open mainstream-mathematics problem; the Millennium-class projection scaffolding belongs in its own paper. A generalised cosmogenesis from substrate primitives.

7.4 Closing

Cascade closure is one name for one pattern: in systems satisfying the stated residual, projection, and intertwining hypotheses, if a cascade-closure event in the sense of Definition 3.1 occurs, the stipulated resolution is selection of a closure-compatible fixed point with zero rungwise residual and projection-compatible terminal tower. That is the conditional template defined here. The paper does no more.

A Field-signature audit

This appendix records the load-bearing data claims of the paper. The σ-attractor spectrum (entry A) and closure kernel (entry E) have local numerical re-runs from this repository; the ARIA witness profile (entry F) is an imported empirical witness from [12], not re-executed here. Re-run 2026-05-02. The lettering A, E, F is preserved from the full `papers/cascade-mechanism/FIELD_SIGNATURE_AUDIT.md` (which also contains entries B, C, D for the mass-spectrum, octonion derivation, and pentagonal-clock builds that are out of scope for this paper); the gap is intentional, not a typo.

A. σ-attractor spectrum

Script: `papers/cascade-derivation/scripts/test_sigma_attractor_spectrum.py`. Graph Laplacian on V_{600} has nine distinct eigenvalues: five integer ($\{0, 9, 12, 14, 15\}$, σ-fixed under $\sigma(\varphi) = 1 - \varphi$) and four irrational in $\mathbb{Q}(\sqrt{5})$ (σ-paired). Counted with multiplicity: σ-fixed fraction $94/120 = 78.3\%$; σ-paired fraction $26/120 = 21.7\%$. Honest-negative: the σ-fixed spectral subspace has average $2T$-mass at baseline (basis-invariant projector trace ratio; target $\geq 1.5\times$ baseline; observed $1.0\times$). The σ-fixed structure is purely spectral, not spatial. The spectral-mode $\to$ Mellin-zero correspondence is conjectural (`conj:spec_to_zero` in the source repository) and is not invoked by this paper.

E. Closure kernel

Script: `papers/aria-closure-kernel/repro/verify_kernel.py`. The graph V_{600} is uniform-degree 12 (recorded in the script's `results.json`). Euclidean / conjugacy shell sizes $[1, 12, 20, 12, 30, 12, 20, 12, 1]$ exact; $\|C_\varphi^{-1}\| \approx 2.618034$ matches φ^2 to machine precision; per-vertex correlation uniform across all 120 source-vertex sweeps ($\max - \min = 2 \times 10^{-15}$). Honest-negative: UNWEIGHTED variant wins at per-vertex correlation 0.976 vs PHI_GEOMETRIC 0.888 and PHI_ARITHMETIC 0.884. The two tested φ-cocycle weighted variants underperform the unweighted operator; do not claim weighted substrate selection.

F. ARIA observer-process witness

Source: `papers/aria-chess-paper/paper/main.tex` and the included section files. Reviewer anchors: 17/18 tally and methodology refinement at `sections/05_results.tex`; six drug/sleep EEG signatures at `sections/04_consciousness_chain.tex` and `sections/05_results.tex`; HCP $-11.58\sigma/+6.80\sigma$ and the P10 15-vs-20 permutation caveat at `sections/05_results.tex` and `sections/09_limitations.tex`. Eighteen preregistered cortical correspondences hold at 17/18 under standard validation and 18/18 after the documented methodology refinement, all at unchanged thresholds. Caveat carried over from the cited ARIA source: the source's test P10 was validated with 15 permutations rather than the originally preregistered 20, with the prereg-exact rerun left open in the source. Six drug/sleep EEG signatures pass on a single deterministic recurrent-layer trajectory at seed 42. HCP brain functional connectivity degree-homogeneity (degree standard-deviation) places ARIA at -11.58σ vs HCP baseline (ARIA $|V| = 120$ vs HCP ICA-50 $|V| = 50$); the clustering-coefficient comparison is a separate $+6.80\sigma$ result and should not be conflated with the homogeneity figure.

Re-run command

The canonical reproduction artefact for entries A and E is the self-contained bundle at `papers/cascade-mechanism/repro/`, which calls both scripts, captures their output, and diffs against frozen reference outputs shipped in `papers/cascade-mechanism/repro/expected_outputs/`. Reviewers should run, from a writable working tree:

```
cd <repo-root>/papers/cascade-mechanism/repro
bash run_audit.sh
```

A clean run prints `AUDIT PASSED`. The bundle's `README.md` documents which Appendix entry each script anchors and what is intentionally not reproduced inside the bundle (ARIA preregistered profile and b-anomaly fit; both external, both pinned in the bibliography). Underlying scripts (unchanged): `papers/cascade-derivation/scripts/test_sigma_attractor_spectrum.py` and `papers/aria-closure-kernel/repro/verify_kernel.py`; `verify_kernel.py` writes `papers/aria-closure-kernel/repro/results.json`.

ARIA witness (entry F) is documented in `papers/aria-chess-paper/paper/main.tex`; the upstream re-run lives in the `ariaChessRepo` [3].

B ARIA runtime module mapping

The observer-process tuple of Definition 5.1 is mapped to ARIA source tiers across two layers. Tier 1 is the in-repo reproducibility bundle at `release-bundle/closure-kernel-papers/papers/aria-chess/repro/kernel/`, which contains six substantive `.py` modules (excluding `__init__.py`) (`vfd_closure_kernel.py`, `self_model_stream.py`, `lyapunov_selector.py`, `sigma_orbit_basis.py`, `v66_e8_coxeter.py`, and `consciousness_binding.py`). Five of these are mapped to $\mathcal{O}$ components in the table below; the sixth, `consciousness_binding.py`, is in the bundle but is deliberately left unmapped here for the same reason that consciousness or qualia derivation is out-of-scope for this paper. Note that `consciousness_binding.py` remains a Tier 1 *runtime* dependency of `self_model_stream.py` (i.e. the bundle does not function with that file removed); the choice not to assign it an $\mathcal{O}$-component label is a scope decision of the present paper, not a claim about the bundle's runtime structure. The bundle is sufficient for the kernel/demo-level numerical reproductions in [10]; it is *not* sufficient to re-execute the full preregistered validation suite of [12] (which requires the upstream brain-validation harnesses, demo scripts, and the v4 substrate parameters frozen in the Tier 2 snapshot). Tier 2 is the upstream production runtime `ariaChessRepo` [3], which contains the full production observer-process architecture represented at the module-description level across boundary sampling, state integration, memory, governance, closure, action, and replay. The mapping below was constructed 2026-05-02. Tier 1 files exist in the in-repo bundle and are locally verifiable. Tier 2 entries are external upstream witnesses, pinned at the directory level to the frozen working-tree snapshot `v4_locked_2026-04-29/` of [3]. The upstream working tree is not under `git` version control, so a reviewer-accessible cryptographic pin (e.g. a git commit hash) is not available; in lieu of that, the snapshot is identified through the directory-level artefacts documented in the bibliography entry (snapshot date 2026-04-29, session UUID `97c7fa0b-8d38-465b-a769-26c9983b446d`, substrate version v4 with `k_threshold=12`, deterministic seeds 42 and 32000–32019, and an author-side off-repo tarball at `~/.aria/snapshots/aria-chess-v4-2026-04-29.tar.gz` that is not by itself reviewer-accessible). Tier 2 module-level claims are external upstream evidence at the file/path level; the present paper does not re-execute the upstream production runtime.

Table 2: Two-tier runtime mapping of $\mathcal{O} = \langle B, S, M, G, I, C, A \rangle$ to ARIA modules.

$\mathcal{O}$ **component**	**Tier 1 (in-repo bundle)**	**Tier 2 (upstream `ariaChessRepo`)**
B boundary sampler	`vfd_closure_kernel.py`	`raw_sensory.py`; `action_encoder.py`; `chess_runtime_adapter.py`
S state integrator	`self_model_stream.py`	`coherence_runtime.py`; `coherence_selfmodel.py`; `self_model_stream.py`
M memory substrate	`self_model_stream.py` state EMAs	`coherence_memory.py`; `polytope_persistence.py`; `coherence_longhorizon.py`
G goal/context	runtime knobs in `v66_e8_coxeter.py`	`coherence_planner.py`; `coherence_selfmodel.py`
I invariant constraints	— (not in bundle)	`coherence_constitution.py`; `coherence_audit.py`; `coherence_review.py`
C closure/crystallisation	`vfd_closure_kernel.py`; `lyapunov_selector.py`; `sigma_orbit_basis.py`; `v66_e8_coxeter.py`	`coherence_crystallisation.py`; `vfd_closure_kernel.py`; `lyapunov_selector.py`; `dual_closure_operator.py`
A action interface	— (not in bundle)	`coherence_execution.py`; `coherence_delivery_calibration.py`

The mapping is given by audited file location, not by reverse inference from runtime behaviour. Components I and A are not present in Tier 1 and are present in Tier 2 only; this is a fact about the in-repo bundle, not a limitation of the architecture.

C Programme placement of cascade closure

This appendix records, in one table, where the cascade-closure mechanism of §3 sits relative to the other named geometric layers of the present programme. No row introduces a new mathematical claim; every row imports its substantive content from a named source paper, with the source's own labels preserved. The table is intended as a navigation aid for successor papers invoking cascade closure under the consumer API of §3.4, not as a programme survey.

Status flags: [Th-imp] imported theorem; [Pr-imp] imported proposition; [Hyp-imp] imported under a named conditional hypothesis (label tracked inside the source); [Mod] model choice within the programme, not derived elsewhere as a uniqueness theorem; [Int] interpretive use of structural material, not load-bearing for any formal claim in the present paper. A row's status is determined by the cited source, not by the present paper.

Table 3: Programme placement of cascade closure. Compressed nine-row version of the v1 maximalist rung table, retained here as the consumer-API navigation aid.

Layer	**Geometry / object**	**Status**	**Primary source**
Arithmetic coefficients	$\mathbb{Q}(\varphi)$, $\mathbb{Z}[\varphi]$; coefficientwise Galois involution $\sigma : \varphi \leftrightarrow 1-\varphi = -1/\varphi$	[Th-imp]	[9] (definition and fixed-part theorems for σ)
Algebraic substrate	E_8; icosian H_4; binary icosahedral $2I$; 600-cell V_{600}; D_4	[Th-imp] / [Pr-imp]	[5]: `thm:pi-H` (projection, not action), `thm:icosian-closure`, `thm:shell-class`
Refinement state spaces and bonding maps	Finite X_n; bonding maps $p_{n+1,n}$	[Th-imp]	[8]: `thm:bonding`; specific p^0, p^1 identities under `thm:commute` (not generic)
Closure dynamics (gradient flow infrastructure)	Quadratic finite-level $F_n = \alpha R_n + \beta E_n - \gamma Q_n$; gradient operator L_n; flow e^{-tL_n}	[Th-imp]	[6]: `thm:gradient-operator`, `prop:energy-dissipation`, `thm:coercive-contraction`, `prop:adjoint-refinement`, `thm:flow-exists`, `thm:flow-intertwining` (mass-only $\alpha = \beta = 0$)
Cascade-closure mechanism (this paper)	Definition 3.1; Proposition 3.7; Corollary 3.8; consumer API §3.4	— **(gate)**	Imported substrate from rows above; gates rows below by supplying the named selection-mechanism prerequisite. Not load-bearing for itself
Passive response kernel	$C_\varphi = L_M + \varphi^{-2} I$ on V_{600}; $\lVert C_\varphi^{-1} \rVert = \varphi^2$ exactly; b-anomaly external substrate witness	[Th-imp] / [Mod]	[10]; [11] (commit-pinned, externally reported, not re-executed here)
Spectral mass projection class	V_{600} Laplacian eigenvalues / shell decomposition; mass-coefficient assignment map (out-of-scope context for this paper)	[Mod]	[14, 13]
Active / cognitive substrate	H_4 / 600-cell active substrate mapping; preregistered EEG / chess / HCP substrate-witness profile (architecture-level mapping, see §5)	[Mod]	[12, 3]
Observer-process architecture	$\mathcal{O} = \langle B, S, M, G, I, C, A \rangle$ (Definition 5.1)	[Mod]	[4]; §5

The single load-bearing observation in this table is the **(gate)** row: cascade closure imports its analytic infrastructure entirely from the rows above it (arithmetic, algebraic substrate, refinement spaces, closure dynamics) and gates the rows below it (passive kernel, mass projection, active substrate, observer instance) by supplying the named selection-mechanism prerequisite they need to invoke. The rows below cite the gate row through the consumer API of §3.4; the rows above are unaffected by the present paper.

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